Data and AI Statement

Meiyu Prepare uses public data and local statistical-learning code to study an offline method for rainfall reference events and to explain stock-relocation preparation costs for small shops. It is not an official warning service, a shop-flood predictor, or a loss-probability product.

> **Current status.** The project has not completed real merchant validation, measured avoided loss, demonstrated revenue impact, or demonstrated carbon reduction. No equipment was purchased, no merchant or school back-office data was connected, and no personal information was collected. Cost examples are hypothetical scenarios in CNY.

Data sources

Source	Use	Main limitation
NOAA SSODv2	Archived station-reference information used to explain historical labels and completeness in the English pages	Daily summaries may under-report; complete codes are not high-precision ground truth and are not a live station feed in the English pages
NOAA SSOD format and ISD format	Check day boundaries, units, AA1-AA4, and report codes	Excluded cases lack conservative complete 24-hour truth
GHCN-Daily	Urban Hangzhou cross-reference	Different station; row-level boundary unverified and not merged
Open-Meteo Previous Runs	ECMWF IFS 0.25-degree fixed-lead daily sequence	Aggregation can combine different initializations; one decision-time availability unverified
Open-Meteo Forecast API	Button-triggered **Forecast** in the original Python/Chinese research workbench (`app.py`)	Gridded numerical forecast, not an observation or ensemble source
Open-Meteo Ensemble API	Automatic **Ensemble** in the English main app (`/en/`); button-triggered **Ensemble** in the English static historical demo (`/en/demo.html`)	Members and fractions are uncalibrated numerical samples, not shop-flood or loss probabilities
Open-Meteo Historical Weather API	Archived ERA5 precipitation and temperature context shown for explanation in the English pages	Reanalysis is not a station observation, is not a complete real-time English-page implementation, and is not used to fill labels
Public Hangzhou and Zhejiang materials	Problem context and preparation checklist	No merchant interview, intent, or payment validation

Complete URLs, byte hashes, retrieval times, and coordinates are in `data/source_manifest.json`; local source links and scope notes are in `docs/action_sources.md`.

The 2026 same-station check is separate in `external/hangzhou_2026`. Of 273 January-September configured days, 85 are complete paired days and one is an event. Complete reference reaches only January-March; April-September lacks complete references. We do not claim 2026 plum-rain validation.

Historical experiment

The historical experiment uses 636 paired days: 276 training, 120 validation, and 240 test days, with 18 test events. The event threshold is 25 mm of nominal UTC-day reference precipitation, a research threshold rather than an official warning level.

The AI task maps numerical-weather inputs to the event probability. The selected method is a small logistic regression using `logistic_rain`, with `C=10` selected only on validation. Test Brier is 0.0455 for the selected model and 0.0697 for the fixed training-period event-rate baseline. A temperature/humidity comparison scored 0.0444 but was not selected. Lower Brier is better.

The model is frozen offline. It does not retrain in the browser, choose a test threshold, or create a source-free prediction. The current-weather page is a separate public forecast display, not the historical model running online.

The original continuous-rain rule and selected model share ROC-AUC 0.9712. The model adds probability expression but not new rainfall-ranking ability. Validation has three events and test has 18, so some probability bins under-estimate event frequency. The raw 25-mm rule's 0/1 squared-error score remains 0.0625.

Online weather and cache

Online weather and the frozen experiment are separate. Python cache under `runtime/weather` stores raw response, content hash, retrieval time, and source-specific completeness; the browser stores a local cache. Forecast older than three hours, or background/station reference older than 24 hours, is stale. Retrieval time is not initialization or provider publication time.

- The English main app (`/en/`) automatically requests Hangzhou **Ensemble** weather and reuses a valid browser cache.
- The English static historical demo (`/en/demo.html`) keeps history offline and requests **Ensemble** only after the user clicks refresh.
- The original Python/Chinese research workbench (`app.py`) keeps its **Latest weather** page button-triggered and uses **Forecast**.

Timeouts, rate limits, wrong units, wrong station, duplicate timestamps, malformed response, and missing hourly precipitation do not replace a valid cache. A 24-hour window is not assigned a complete total when any hour is missing and is never filled with zero. Ensemble fractions are uncalibrated internal frequencies, not shop-flood or loss probabilities. Forecast API cannot be presented as ensemble.

The delivered snapshot is labeled with its actual capture time. The 2026-10-19 snapshot must not be presented as freshly updated when a page is opened offline. ERA5 and SSOD shown in the English pages are archived explanatory information, not a claim that those pages implement a complete real-time station-and-forecast service. The English main app's automatic **Ensemble**, the English static demo's button-triggered **Ensemble**, and the original workbench's button-triggered **Forecast** are distinct behaviors.

Reproducibility references

Frozen model and score files are `reports/evaluation.json` and `reports/experiment_run.json`. Source manifests and public snapshots remain in the repository. English materials were prepared on 2026-10-24; historical and capture dates remain unchanged.

AI assistance and human responsibility

Codex assisted with public-source lookup, parsers, tests, app implementation, and material drafts. The code uses Python, pandas, NumPy, scikit-learn, Streamlit, and Plotly. This stage does not require an OpenAI API key; language-model prose is not rainfall data, labels, or a weather decision.

The human team must verify identity, contacts, form fields, user assumptions, links, and final claims. Names, organization, and email are left for the team; none are invented here.

Promotional media

The current Chinese film is 73.866666667 seconds, 1920x1080 at 30 fps, 2,216 frames, and 10 scenes. Frames were generated programmatically and encoded with WebCodecs; screenshots, screen recordings, and external image/video inputs were not used as visual inputs. It has one AAC track and no narration.

Music is **Young Visionaries - Eric Matyas**, used under the **Soundimage International Public License**, with attribution `Music by Eric Matyas / www.soundimage.org`. Source: soundimage.org/techno. License: Soundimage license page. The source excerpt is 17.313872945628-91.180539612295 seconds at original speed and pitch, continuously cut with fades and no loop. Source files and third-party notices are under `media/latest/source/`.

The separate English cut is `media/en/meiyu-prepare-en.mp4`: 87.066666667 seconds, 1920x1080, 30 fps, 2,612 frames, 10 scenes, and 132 beats, with one AAC track and no voice-over. Its source excerpt is 6.764338247872-93.831004914539 seconds at original speed and pitch. The current file size and SHA-256 are recorded in `media/en/video-info.json` and `docs/en/video-technical-check.json`. Independent visual and technical checks for the current English cut are recorded there; full team viewing/listening remains pending. See README.en.md for verified publication status; the Chinese and English media are separate assets.

Use and licensing

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> **Safe-use boundary.** The project has not shown reduced real-world damage, increased operating income, avoided loss, or carbon reduction. Outputs are not official warnings, emergency instructions, or shop-flood forecasts. The climate value is a preparedness question requiring later pilot evidence.